

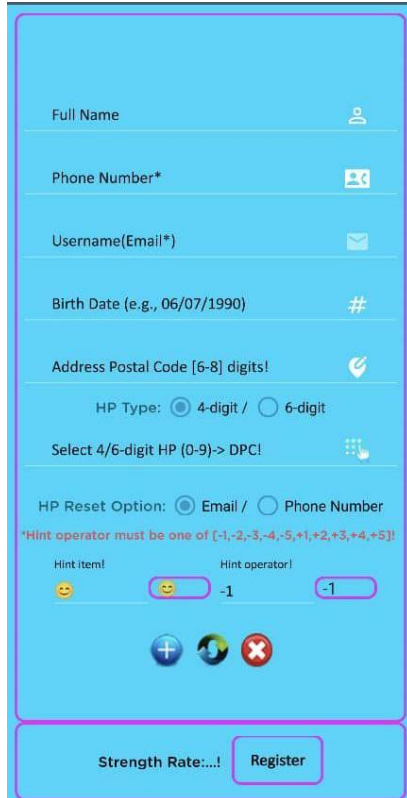
Dear Participant,

We appreciate your participation in this software testing survey, which will greatly help to improve our project considering the public opinion. Before installing the HDynPIN Android app, please read the following guidelines:

Download link of Android App: <https://github.com/GoldenAL1990/HDynPIN-Protocol/blob/main/HDynPIN-Protocol.apk>

The HDynPIN app consists of two pages (Registration and Authentication). In the following, you can see the screenshots of these two pages. *Note that our experiments confirm the security robustness level of HDynPIN is three times higher than standard PIN scheme and outperform other state-of-the art protocols against active adversarial attacks inferring a victim device. According to our comparative results, the attack robustness rate for HDynPIN is (5/5), and for the Standard PIN scheme is (2/5).

(a) Registration (or Sign-up) UI



(b) Authentication (or Sign-In) UI



The difference between the HDynPIN and other existing PIN-based schemes is that user has to pick a hidden PIN (HP) and at least a set of three hint items and operators paired with each other that must be non-repetitive items which could be any symbols (e.g., words in different languages, emojis and so on). These settings help the user to enter a one-time valid dynamic combination (DPC) during the authentication. A user must take the following steps.

- 1) The identity information is chosen and requested to make sure that the user does not pick a part of her/his public available data that may lead to cracking the PIN based schemes by brute force attacks.
- 2) Email/Phone number: The HDynPIN also requires a user to set email address or a phone number for resetting the settings in case of forgetting the HP. Once it is verified, the user is allowed to find the DPC.
- 3) HP -> DPC: User has to set and memorize a 4-digit or 6-digit HP (e.g., 3619) during the registration (sign up) that will be used to find the DPC according to a one-time valid hint item.
- 4) AI algorithm will generate a one-time valid hint item (e.g., +=1) and a randomized entry pathway (e.g., one spot is checked randomly for 4-digit entry and two spots are checked randomly for 6-digit entry). Then the user can see the hint item to remember the corresponding operator or tap on it two times to see the corresponding operator. Hence, the DPC (e.g., 47[checkmark]0) can only be verified successfully once.

* Note that all the list of hint items and operators can be defined by user during the registration phase and the AI only ensures that they are not repeated for at least three authentication processes.